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ORIGINAL PAPER

Implementation of a Peer-Mediated Intervention to Teach Behavioral Expectations for Two Students on Autism Spectrum and a Student with ADHD in an Inclusive Elementary Classroom in Taiwan

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Abstract
One of the characteristics of individuals on the autism spectrum is the restricted, repetitive patterns of behaviors, interests, or activities which have been considered the result of a direct deficit in self-regulation skills. Peer-mediated intervention (PMI) is one of the ways to support young children on the autism spectrum for following school routines independently. PMI is an evidence-based practice based on behavioral principles to train typical peers to help with students on the autism spectrum. By using PMI, students on the autism spectrum can have more interaction opportunities with typical peers. Students on the autism spectrum can become less dependent on adult support to complete the expectations independently or with peers' support. The aim of this study is to use PMI to support students with autism and ADHD aged 7–8 in a regular classroom to complete the behavioral expectations in Taiwan. This study used a multiple probe design across three participants to determine the effects of the intervention. The researchers trained seven typically developing peers on how to prompt and reinforce the target children. This study took both peers and target children's data. The data shows a positive result for target children to follow the expectations independently and most peer models can prompt and reinforce the target children after being trained. Our study has extended the effectiveness and usage of PMI in teaching behavioral expectations in Taiwan.

Keywords Autism spectrum disorder · Peer-mediated intervention · Inclusive education

Introduction

This study aims to identify the effect of PMI on supporting the self-regulation of children on the autism spectrum and with Attention Deficit and Hyperactive Disorder (ADHD) to complete the behavioral expectations of routines in an inclusive elementary classroom in Taiwan. Individuals with ADHD experience difficulty paying close attention in completing school activities or routines (CDC, 2021). Individuals on the autism spectrum have the characteristics of restricted, repetitive patterns of behaviors, interests, or activities; and persistent deficits in social communication and social interaction across multiple contexts (American Psychiatric Association [APA], 2013; Centers for Disease Control and Prevention [CDC], 2021). Those characteristics directly affect their self-regulation skills in following behavioral expectations (Barnard-Brak et al., 2014). Given that peer mechanisms place an important role in self-regulation development in early childhood (Pahigiannis & Glos, 2020), it is necessary to examine the effectiveness of peer-mediated intervention (PMI) in increasing children's self-regulation skills. Moreover, PMI can stimulate interactions between children with and without disabilities which aligns with the current trend and law for inclusion in Taiwan.

Autism Spectrum Disorder, ADHD and Self-Regulation Skill

The 2020 Special Education Statistics Annual Report published by the Taiwan Ministry of Education (2020) points out that there is an increasing number of students with disabilities. According to that report, the third majority population is the autism spectrum disorder (ASD, 17.23%) totaling

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7,459 children with autism spectrum. Students on the autism spectrum have the characteristics of restricted, repetitive patterns of behaviors, interests, or activities (APA, 2013; CDC, 2021) which have been considered the result of a direct deficit in self-regulation skills (Barnard-Brak et al., 2014). Besides, individuals with ADHD also have self-regulation deficits in completing teacher expectations or school activities (Barkley, 2011).

Self-regulation, which includes emotional and behavioral control, is a crucial skill in early development that impacts the whole life (Pahigiannis & Glos, 2020). One of the ways to reflect children's self-regulation skills is whether they can follow the behavioral expectations of classroom routines (Carter & Pool, 2012; Wahman & Anderson, 2021). Children with limited self-regulation skills usually come with challenging behaviors and low social and emotional skills which cause a high likelihood of experiencing difficulties in their later school years, including academic failure, safety issues, peer rejection, and negative relationships with teachers (Carter & Pool, 2012; Jones et al., 2015; Pahigiannis & Glos, 2020; Wahman & Anderson, 2021). Additionally, children with low self-regulation skills have a higher possibility to emit challenging behavior which can impact all aspects of the classroom environment (Carter & Pool, 2012).

Previous research has focused on children with disabilities of self-regulation in following the expectations of routines with a variety of interventions. Goetz et al. (1983) targeted the clean-up routine after free play. They reported that preschoolers reduced their time on putting blocks away by pairing teacher praise and auditory stimulus (i.e., a click sound in a 20-s schedule). Connell et al. (1993) used a self-assessment and teacher praise to successfully teach transition skills for preschoolers with developmental delays. Specifically, the researchers put happy or sad faces adjacent to a booklet immediately after the transition, and the child marked the face representing their self-performance of that behavior. Davis et al. (2000) reported that using high-probability requests (e.g., "Touch head, touch shoulder, raise your hand, go to the classroom") and a preferred item can increase successful transitions in children with behavior problems. Ferguson et al. (2004) found that physical prompting, audio prompt (i.e., the bell rang), and edible reinforcement were an easily implemented and effective set of strategies to reduce the amount of time between transitions in a self-contained preschool classroom. Wahman and Anderson (2021) reported a positive effect for three preschoolers with disabilities to follow the expectations of naptime and circle time transition routines after training teachers to provide specific praise, conduct social stories, and role-play with the situations of routines. However, previous research has little focus on the peer factor in training self-regulation skills. Pahigiannis and Glos (2020) pointed out that the perspective of peer mechanisms places an important role in self-regulation development in early childhood after reviewing 20 studies related to peer influence and self-regulation skill training. They urged that future research should focus on peer process as a universal prevention intervention for children with disabilities to self-regulate.

Peer-Mediated Intervention (PMI)

Peer-mediated intervention (PMI) using behavioral principles to train typically developing children to support children with disabilities has been recognized as an evidence-based practice by the National Professional Development Center on Autism Spectrum Disorder (NPDC) (Odom & Strain, 1984; Banda et al., 2010; Sam et al., 2020). Furthermore, PMI on self-regulation development is based on the theoretical framework of Vygotsky's social development theory and Bandura's social learning theory (Pahigiannis & Glos, 2020). PMI is based on Vygotsky's theoretical framework in which children build up their social skills by interacting with other individuals and the environment (Vygotsky, 1978; Tzuriel & Shamir, 2007; Simpson & Bui, 2016). PMI is based on Bandura's social learning theory in which children with disabilities can observe, model, and imitate the typical peers' behaviors to learn the specific behaviors in each social context (Bandura & Walters, 1977; Barber et al., 2016; Owen-DeSchryver et al., 2008).

PMI's effectiveness has been examined in a variety of domains such as social communication and interaction skills (Harper et al., 2008; Jung et al., 2008; Banda et al., 2010; Kamps et al., 2015; Katz & Girolametto, 2015; Hart & Banda, 2018; Dueñas, et al., 2021a), play skills (Hu et al., 2018; Maich et al., 2018; Dueñas et al., 2019, 2021b; Covey et al., 2021), academic skills (McCurdy & Cole, 2014; Haas et al., 2020) and physical education (Ward & Ayvazo, 2006). Having typical children as a peer model can support the children on the autism spectrum to generalize their skills to a different setting and peer models can serve as a cue for target children to maintain their skills (McCurdy & Cole, 2014; Dueñas et al., 2019; Kent et al., 2020; Boudreau et al., 2021). Peer inclusion also has a positive effect for children with ADHD. Cordier et al. (2018) discussed 17 studies that use peer inclusion on children with ADHD in their systematic literature review. They concluded that peer mediation is an appropriate method for students with ADHD to improve their social skills and generalize those skills. Moreover, Harper et al. (2008) stated that having peers provide support is more naturalistic and more motivating than if adults would have implemented the same procedures in a school setting. Furthermore, teachers can save time to focus more on child safety and teaching while children on the autism spectrum can still finish their work with peer support (McCurdy & Cole, 2014). By doing so, it decreases the target children's dependence on adults' help and gradually

increases their independence (McCurdy & Cole, 2014; Covey et al., 2021).

Not only children with disabilities but also the typical peer models can benefit from the PMI program. Peer models who are socially connected to children with disabilities can benefit from maintaining a strong and positive role in the classroom (Locke et al., 2012; Kent et al., 2020; Travers & Carter, 2021), gaining leadership skills (Bene et al., 2014; Harris et al., 2009), improving their self-efficacy (Corbett et al., 2014; Kent et al., 2020), achieving pragmatic language capacity and performance (Parsons et al., 2020), and having enjoyment interacting with children with disabilities (Travers & Carter, 2021). Nevertheless, PMI research should address peer models' benefits with caution and focus more on peer models' benefit by collecting data on their behaviors, understanding their perceptions in participating in the PMI program, monitoring their changes during and after the intervention, and making adjustments accordingly (DiSalvo & Oswald, 2002; Schaefer et al., 2016; Travers & Carter, 2021).

Fortunately, a few previous studies have examined the effectiveness of PMI in self-regulation skills of following behavioral expectations. Carden Smith and Fowler (1984) demonstrated that the peer-monitored token economy is effective in reducing students' disruptive behaviors during the classroom routines of transition time. In addition, Sainato et al. (1987) targeted three classroom transition times (i.e., circle to the lesson, a snack to the bathroom, and circle to language) for independent movement and appropriate behavior of preschool children on the autism spectrum. They found that peer buddies and visual cards for all children can serve as a specific direction for children in transitioning to another area. Similarly, Harper et al. (2008) used visual cue cards in peer training to facilitate a more structured training session. They used it as a tool for peers to successfully gain attention from children on the autism spectrum. Fudge et al. (2008). reported that the color wheel system can effectively increase on-task behavior for every student in a second-grade classroom during in-class transitions. To be more specific, they taught their students to follow different rules based on different colors; for example, the color green means to use the inside voice to share with others, respect others, and hands and feet to self. However, limited to no studies examine the effect of PMI in supporting the self-regulation skills of children with ADHD or on the autism spectrum in following the behavioral expectations of routines in an inclusive early education setting.

PMI and Inclusive Education in Taiwan

The 2020 Taiwan Special Education Statistics Annual Report shows a growing number of students with disabilities who receive education in a regular classroom with typical peers. That is to say, students with disabilities spend more time in a regular classroom with more time to stay with their typically developing peers. However, students with characteristics of social and communication deficits may face social difficulties in an inclusive setting (APA, 2013). PMI facilitates the interaction opportunities between children with and without disabilities which fulfill the key indicators of implementing inclusive education in Taiwan (Huang et al., 2017; Travers & Carter, 2021).

Some research has been implemented in Taiwan to examine the effect of PMI in specific skills such as play and social skills (Yang et al., 2003; Feng & Feng, 2005), life skills (Chuang et al., 2013), academic skills (Lu & Wang, 2000; Tsuei, 2006; Feng et al., 2018), and speech-generating devices (Wu et al., 2020). Wu et al. (2020) trained nine typically developing peer tutors to teach science concepts and model the usage of speech-generating devices following a script to three elementary school students with significant intellectual disabilities aged 8–9 in Taiwan. With multiple baseline designs across participants, Wu et al. reported that the PMI is effective in improving participants' targeted science knowledge, increasing communication interactions between peers with/without disabilities, and increasing the usage of speech-generating devices as a different communication mode during the science experiment activities. For play and social skills, Yang et al. (2003) used A-B design and semi-structured interviews with parents and peers to examine the effect of integrated playgroup models in increasing cognitive/symbolic play and social play for two students on the autism spectrum. Their results showed that each child grew their abilities in reciprocal social interactions and symbolic/pretend play after participating in playgroup. Similarly, Feng and Feng (2005) focused on social interaction skills. They used multiple probe designs across peer-trainers to examine the effect of peer-implemented pivotal response training for children on the autism spectrum. Four 6th grade typical peers and one child on the autism spectrum participated in the study. They reported that the frequencies of maintaining interactions and initiation actions increased and those target behaviors were extended to an untrained peer. Other than play and social skills, Chuang et al. (2013) primarily targeted life skills and secondly focused on peer acceptance. They used the A-B-A reversal design and reported that peer tutoring (four 5th graders) helps elementary school students with intellectual disabilities (one 1st grade and one 3rd grade) increase their life skills. They reported that peer acceptance of students with special needs increased by using C statistics and descriptive analysis. For academic skills, Tsuei (2006) used peer-assisted learning strategies on 69 third-graders, including two students with learning disabilities for 12 weeks. Tsuei reported that students in the experimental group had a better performance than the control group on Chinese language achievement tests and reading comprehension tests. Also,

the students with learning disabilities improved their Chinese language skills at higher rates than general education students in the same classroom. Not only language skills but also peer relationships were measured. Interestingly, Tsuei reported that it was an effective intervention to only enhance higher-ability students' peer relationships, but there was no such effect on general-ability students, low-ability students, and two students with learning disabilities. Feng et al. (2018) also focused on increasing students' Chinese learning using peer mentoring. Different from Tsuei, Feng et al. focused on 24 eighth graders (12 dyads) without disabilities. They reported that peer mentoring is an effective intervention to increase students' Chinese acquisition by utilizing observation, interviews, and questionnaires. According to the above PMI literature review in Taiwan, few studies examined the effectiveness of PMI for students with disabilities, let alone the usage of PMI in supporting children with disabilities to follow behavioral expectations. There is no study focusing on implementing PMI based on applied behavior analysis (ABA) principles in supporting children with disabilities to follow the behavioral expectations in a regular classroom in Taiwan.

Research Gap and Opportunities

Despite the high volume of interventions targeting PMI, few have intentionally measured using PMI in supporting children with disabilities targeting self-regulation development in following the behavioral expectations of routines in early childhood, and nearly no study has done so in Taiwan. Moreover, not much PMI research collects data on peer models' behaviors and perceptions in supporting children with disabilities. As such, it remains largely unknown how PMI may influence children with disabilities in self-regulation skills, and peer models' advantages and opinions in having a PMI program in an inclusive classroom in Taiwan.

To address these gaps in the literature base, this study expands the field of PMI in five primary ways. First, previous studies have examined the usage of PMI primarily on play and social communication skills, and academic skills. There are limited previous studies that used PMI in teaching self-regulation skills. This study expands the usage of PMI in teaching self-regulation for following behavioral expectations of routines by systematically replicating the study of Wahman and Anderson (2021) that identified teacher praise, and social stories with role-playing were effective to support preschoolers with disabilities to demonstrate self-regulation in following expectations of classroom routines. Secondly, there is a paucity of studies examining the effectiveness of PMI based on the ABA principle in Taiwan. Our study can enrich the number of PMI studies that took place in Taiwan. Third, our study reports the peer model's treatment integrity and social validity data which is connected with the suggestion from the current literature review to focus on peer models' impact (Travers & Carter, 2021). Fourth, our study examines the extent to which the intervention would be effective for peer models and target students closer in age in a regular elementary classroom which aligns with the suggestions from the current study (Covey et al., 2021), corresponds to the indicators of inclusive education in Taiwan (Huang et al., 2017), and meets the current inclusion trend in Taiwan (Taiwan Ministry of Education, 2020). Last but not least, our study pairs up the students in a strategic way which is also aligned with the suggestions from the current study by teacher referral considering their personalities, willingness to participate, previous history of interacting with target children, regular attendance, and abilities to complete behavioral expectations of routines (Covey et al., 2021).

Research Questions

1. What is the effect of a peer-mediated intervention (PMI) on target participants' behavior during the transition to activities?
2. What is the effect of the peer training on the peer models for assisting the target participants?
3. To what degree did the peers implement PMI as designed?
4. What are the target children's and peer models' perceptions on participating in the PMI program?

Method

Participants

Both observational data from the researcher and referral from the master teacher were needed to be identified as a potential child participant. Ten children in the second or third grade (aged 7–9) were recruited to participate in this study; two students on the autism spectrum, one student with ADHD, and seven typically developing children participated. Each child with a disability (i.e., target child) was matched with two to three typically developing children (i.e., peer models) who were in the same classroom with the target children.

Target Children

The children who followed teachers' expectations lower than 60% of correct responses, and referred by the master teachers were selected and recruited as target participants. The children who have the official diagnosis and identification of receiving special education services were selected

as a priority. The characteristics of the three target children (James, Henry, and William) are summarized in Table 1. All three participants were studying in the regular classrooms with peer models of a similar age and received special education services from the resource rooms.

James was 8 years 7 months old when this study began. According to his individualized education program (IEP), he has been diagnosed with ADHD-no specific type and other specific anxiety disorder since he was 7 years 5 months by the doctor in the hospital. At the same time, he started to take the emotion-related courses in a small group in the sensory integrated emotional center after school. The occupational therapist in that center reported that James has visual sensory and auditory sensory problems, and he engages in problem behaviors when he fails to complete the task. His master's teacher reported that he is usually slower than his classmates in completing the teacher's expectations. She also mentioned that the classmates' attitude toward James is neutral in general; James doesn't have close friends. He is able to have conversations in daily life with others when his emotions are stable. He talks a lot and would like others to give him immediate feedback. His IEP report shows that his teacher has arranged a peer helper sitting closer to James to help him if needed. When he engages in disruptive behaviors or off-seat behaviors, the public school discipline coordinator gives him a time-out (i.e., stand up at his seat). Due to James's increasing frequency and magnitude of aggressive behaviors (e.g., hurting classmates), he stayed home for several days. Therefore, limited data points of his behavior were collected in this study.

Henry just turned 9 years old when this study began. According to his IEP, he was diagnosed with ASD. He has received occupational therapy and speech therapy in the hospital since he was 2 years old. Now, he also receives occupational therapy in his public school. He had attended the private preschool with an itinerant resource program since he was 3 years old. His IEP report shows that he is slower than his classmates for completing teachers' expectations or tasks, and he needs a peer or teacher's reminder to keep pace with the class. His emotion is stable and he is willing to accept direction and guidance from others. He took few initiatives in interacting with peers. Also, his IEP reported that he shows low motivation to do tasks and suggested he needs to gradually build up the ability to do multiple steps.

William was 7 years 10 months when this study started. According to his IEP, he was identified as the category of ASD by the Committee Responsible for Identification and Placement of Gifted and Disabled Students. He received sensory integration training for 3 months when he was 3 years old. He is receiving occupational therapy, physiology therapy, speech therapy and music therapy, and some courses with psychologists. His emotion is stable in general. If he can't get what he wants, he engages in an emotional outburst for a long amount of time. His teacher has arranged a peer helper to support his learning. His IEP pointed out that he has difficulty adapting to the environment; for example, sometimes he cries when he doesn't know what to do or loses his pace with the class. His IEP suggests that he needs

Table 1 Characteristic of the target children with special needs

	James	Henry	William
Age	8.7	9.0	7.10
Gender	Male	Male	Male
Diagnosis	ADHD	ASD	ASD
WISC-IV	97 (PR42)	84(PR14)	(PR8)
Verbal comprehension	95 (PR37)	93(PR32)	(PR8)
Perceptual reasoning	103 (PR58)		
Working memory	94 (PR34)		
Processing speed	94 (PR34)		
Problem behavior rating scale			
Master teacher	ADHD, combined type		
Discipline coordinator	ADHD, impulsive/hyperactive type		
Mother	Not ADHD		
Scales for assessing emotional disturbance			
Master teacher	121 (PR92) at risk for ED		
Discipline coordinator	126 (PR97) at risk for ED		
Mother	109 (PR74) Not ED		
Adaptive behavior	(PR7)	(PR3)	(PR9)

WISC-IV Wechsler intelligence scale for children-fourth edition, *PR* percentile rank, *ADHD* attention deficit hyperactivity disorder, *ED*

someone to hold his hand to transition from one classroom to another. He also needs a person to help him with the go-home routine; for example, holding his hand to walk from the classroom to the school gate.

Peer Models

Each of the three master teachers was asked to nominate two to three typically developing peers as peer interventionists for the children with disabilities. The researcher collected the observational data on the potential peer models who were referred by the master teacher. The criteria for selecting the peers were showing (a) typical language development as described by the teachers, (b) typical social skills as described by the teachers, (c) attendance in the same classroom as target child, and (d) previous interest in interacting with the target child (Katz & Girolametto, 2013), and (e) completed teacher's expectations of routines for 100% of correct responses.

Settings and Materials

The study took place in three different regular classrooms in a public elementary school in Taipei, the capital city of Taiwan. Each classroom contains one master teacher and 28 students. During the peer training session, data were collected in the resource room which contains seven peer models from those three regular classrooms and two researchers from OSU. During the baseline and intervention sessions, data were collected in the regular classroom. All three target children were sitting at the back of the classroom which was the last line of seats in the classroom. Their peer models were not sitting next to the target children, except one of Henry's peer models. Some of their seats were close to the target children and some were not.

The materials used in this study were a visual cue card with the icon of thumbs up and a performance feedback sheet for the peer models. Each target child had their own visual cue card listing their teacher's expectations of the go-home routine. The visual cue card and the thumbs-up icon were both laminated. The visual cue card was around 21×29.7 cm. On the visual cue card, it shows the researcher demonstrating each of the six steps (i.e., stand up, pack up the school bag, lineup, etc.) Under each picture of the steps, each step was written with three to four Mandarin words. Henry's peer model was provided with the performance feedback sheet to increase the peers' treatment integrity. The sheet was around 21×29.7 cm. There was a table with each step of the expectation listed on the left column. The top row was prompt, and praised, and under the item was "Do it on your own! Good," "Do it after teacher reminds," "Remember to do it next time," or "Our friend doesn't need others' help. He does it by himself."

Recruiting and Informed Consent Process

After receiving IRB approval, both observational data on following routine expectations collected by the researcher and the teacher referral were needed to identify potential children participants in this study. The parent consent forms were sent to the parents of the target students and the parents of the peer models. After receiving signed parent permission forms, the researcher obtained assent from the peer models individually and privately by asking their willingness to be trained, providing support to the target children, and understanding that they can stop participating in study at any time without penalty. If the peer models agreed to participate, they provided verbal assent.

Definition and Measurement of Dependent Variables

This study examined the effect of PMI on a number of expectations completed correctly by the target child. Data were collected one to 4 times per week and once per day depending on the participants' schedules.

Number of Steps Completed Correctly by Target Child

This dependent variable assessed the target children's behaviors in completing the number of expectations. The researcher and two other trained graduate students collected the data and interobserver agreement (IOA) data to ensure accuracy. The data collectors squatted down to reduce the possibility of distracting the class and child participants. There were six steps in each target child's classroom (see Table 2). Each target child had six different classroom expectations which were requested from their master teachers. The coded system was (1) a letter "I" which means the target child independently completed the step in the following three conditions. First, if the target child starts the steps/behaviors in 5 s after the teacher's announcement. Second, if the target child starts the steps/behaviors in 5 s after finishing the last step. Only the last step is different than other steps. The last step would be lining up in the hallway, getting into the line in 10 s after the last student walked out of the door, and staying with the class; (2) a letter "P" means the target child has done that step with peer model's help; (3) a symbol "-" means the target child fails to finish the steps even with peer model's prompt; (4) "N/A" is an abbreviation of "not applicable" in terms of no opportunity for the child to do this step. For example, the teacher asked the class to skip that step for a shorter time of the go-home routine.

Table 2 Behavioral expectations of the go-home routine for each participant

	Behavioral expectations	Operational definitions
James	1. Packing school bag 2. Sweep the floor 3. Circle the number 4. Carry on the school bag 5. Scoop up chair 6. Go lineup on the hallway and get into the line	• When the teacher gives an announcement, the child puts his personal belongings and homework into his bag • The child uses a small broom to sweep the floor around his seat. He also can choose to simply pick up some trash and put it into the garbage bin. If there is no trash, the child needs to perform the behavior of checking that is clean. The child should not throw the trash or keep playing with the trash in hand • When the child finishes sweeping the floor, the child walks to the blackboard to circle his number or ask his peers to help him circle it. The child should not clean up the chalk tray at this time, draw the blackboard, or throw the chalks • The child should put on his school bag and scoop up his chair. The sequence of those two steps can be switched sometimes • If there is enough time for the child to catch up the line, the child can walk or run to the hallway to start walking to the school gate with classmates. If it is too late, this step will not have an opportunity to occur
Henry	1. Packing school bag 2 Stand up 3. Scoop up chair 4. Carry on school bag 5. Set desks in order 6. Go lineup on the hallway and get into the line	• When the teacher gives an announcement, the child puts his personal belongings and homework into his bag • The child stands up in his seat. Fail to do this task would be to keep sitting when the teacher tells the class to stand up • The child put the chair close to the desk to make it organized • The child should put on his school bag and scoop up his chair. The sequence of those two steps can be switched sometimes • The child looks at the bottom of the desk and makes sure it is on the line • The child walks to the hallway, keeps in line, walks with his classmates to the school gate • Steps one to five may happen in a different sequence
William	1. Packing school bag 2 carry on school bag 3. Set desk in order 4. pick up trash 5. Put a chair on the top of the desk 6. Go lineup on the hallway and get into the line	• When the teacher gives an announcement, the child puts his personal belongings and homework into his bag • The child should put on his school bag and scoop up his chair • The child looks at the bottom of the desk and makes sure it is on the line • The child should look around his seat and pick trash up if there is one • The child holds chairs with both hands, raises them up, and up them on the top of the desk • The child walks to the hallway, keeps in line, walks with his classmates to the school gate

Percentage of Steps Implemented Correctly by Peer Models for Prompting and Reinforcement

The secondary dependent variable is the percentage of steps implemented correctly by the peer models for prompting and reinforcement. The researcher and two other trained graduate students collected the data and interobserver agreement (IOA) data to ensure the accuracy. There were six steps or expectations in each classroom. For each expectation, the peer models may or may not have a chance to deliver prompts or reinforcements to the target children. For promptings, the peer models may show and point at the visual cue card for the target children to be able to see. The peer models may also use gestural prompts; for example, the peer models pointed at the schoolbag to remind the target children to put books into schoolbag. The peer models could use verbal prompts; for instances, "Hey! Let's go line up!" The peer models may use physical prompts which means use their hands to gently hold the target children's hands to complete the expectations. The peers can use multiple prompts at the same time. The situations and coding system for prompting are described as follows: (1) A letter "I" means the peer model independently completed the prompting after waiting for 5 s observing whether the target children need their help or not. (2) A letter "P" means the peer model has done that step with researchers/graduate assistants' help. For example, if the peer models did not wait and give the target children a chance to perform classroom expectations (e.g., the peer models were approaching the target children to directly use their hands to help put books into schoolbag), the researcher or graduate assistant blocked the peer models' and reminded them that they need to wait and observe first before giving assistance. Another situation is when the peer models did not provide prompts after the target child was struggling or wandering around for five seconds, the researcher provided verbal prompt to the peer models to show visual picture cards or verbal prompts to the target children. (3) A symbol "–" means the peer model fails to finish the steps even with researcher's prompt. For example, the researcher prompted a peer model to show visual cue cards to the target child but

the peer model refused to do so by putting the visual cue cards into a drawer. (4) "N/A" is an abbreviation of "not applicable" in terms of no opportunity for the peer model to provide promptings. For example, the target children finished the expectations within 5 s independently. Another example is when the teacher asked the whole class to skip the specific steps or expectations.

For the reinforcement, the peer models put the visual thumbs-up card on the specific steps of visual cue card as a reinforcement. The peer models were encouraged to add additional social reinforcement to the target children, such as a specific verbal praise "You did great on picking up the trash!" or having a high-five. The situations and coding system for reinforcement are described as follows: (1) A letter "I" means the peer model independently provided reinforcement when the target children completed the expectations by themselves or with their prompts. (2) A letter "P" means the peer model provided reinforcement with researchers/graduate assistants' help. For example, if the peer models did not provide reinforcement when the target children finished the steps after 3 s. (3) A symbol "-" means the peer model fails to finish the steps even with researcher's prompt. For example, the researcher prompted the peer models to put thumbs-up card on the visual cue card, but the peer models refused to do so by putting the cards away. (4) "N/A" is an abbreviation of "not applicable" in terms of no opportunity for the peer model to provide reinforcement. For example, the target children did not complete the steps by themselves or with peers' prompts, then there is no opportunity for the peer models to provide reinforcement. Another situation was when the teacher announced the class to skip the steps, then there is no opportunity for the peer models to deliver reinforcement either.

Experimental Design

A multiple probe design across three participants was used in this study to evaluate the effects of a PMI program on target children's behaviors of completing teachers' expectations of go-home routine. Multiple probe design is a variation of the multiple baseline design in which researchers intermittently collect the baseline data to probe the participant's behavior before intervention starts (Cooper et al., 2019). The experimental conditions are baseline and intervention which shows the functional relation in this study that the dependent variable (i.e., the percentage of completing the go-home routine steps) is reliably produced by specific manipulations of the independent variable (i.e., the PMI program) (Cooper et al., 2019).

Procedures

The procedures include baseline, peer training, and intervention in this study.

Baseline

In the baseline, all 10 participants did not receive any intervention or prompts from the researcher, and no visual cue card and performance feedback sheet were provided. The researchers stayed at the backdoor of the regular classroom which was 5–10 feet away from the participants to observe the behaviors of the target children and peer models. The researchers took the baseline data on the target children's target behavior of completing the six expectations of go-home routine and peer models' behaviors on providing prompts and reinforcements in corresponding to each six of the expectations.

Peer Training

A 40-min peer training session was provided in the resource room during nap time. Before beginning training, all seven peer models were asked by the researcher about their willingness of being trained in the resource room during their nap time. The peer models were trained by the researchers using behavioral skill training which includes instructions, modeling, rehearsal, and feedback (Dib & Sturmey, 2012; Covey et al., 2021). The researcher provided an overview of the PMI program, the importance of having them as helpers, and the value of visual cue cards and giving reinforcement. Next, the peer models were divided into three groups. Children from the same classrooms were in the same group. The researchers modeled (a) how to use the visual cue card to prompt the target child, (b) wait for three seconds before prompting, (c) prompt three times in maximum, and (d) if the target child is not listening, it's fine to stop prompting for this step and wait for the next step to come or end this session. The researcher also modeled how to reinforce and when to reinforce the target child. Next, each peer model had a quick role-play with the researcher. They also observed the other peer models role-playing with the researcher. The researcher provided immediate feedback to the peer models about where they did well or could be done better. Each peer model had met the standard of implementing 80% of steps correctly (i.e., providing prompts and reinforcement to each expectation) three consecutive times. Henry and William didn't join the peer training session. James joined the role-play session when the training was almost over. After he finished his class in the adjacent classroom, he

asked for role-playing as both peer model and target child with his peer models.

Intervention

The PMI program took place 1–3 times a week for 3–4 weeks. The duration of the PMI intervention for each time was around 5–10 min, which began with the teacher's whole-class announcement of the go-home routine and ended with the target children started heading to the school gate.

During the intervention, the researchers observed the peer models and the target children's behaviors and collected data on them at the same time. When the teacher gave an announcement for the go-home routine, the researcher provided the peer model with the visual cue card and stayed close to the peer model for around 5 feet. If the peer model did not prompt the target child by showing and pointing at the visual cue card with a verbal prompt or did not provide reinforcement, the researcher used a verbal prompt (e.g., "Show him the picture and tell him what's next.") and/or provided a gestural prompt to the peer model. At the beginning of the sessions, the researcher added modeling prompts to Henry's peers because they need more help. The researcher kept modeling with the peer model only, rather than with the target child. The target children did not receive direct assistance from the researchers. Only the peer models received prompting from the researchers to help the target children if needed.

During the intervention, the peer models took turns to help the target children, so there was one peer model that helped one target child. When the teacher gave the announcement of the routine, the peer model finished their own things first and then walked to the target child to provide prompts and reinforcements. If the target child kept sitting and doing nothing or wandering around, the peer model did not directly help the target child finish their job. The peer model tried to get his attention, showed him the visual icon by pointing at the specific expectation, and provided the verbal prompts by saying "Stand up," "Put your belongings in your schoolbag," "Put on your bag," "Put the desk in order," "Pick up the trash on the floor," "Go lineup!" If the target child completed the expectation individually or with the peers' prompts, the peer model immediately put a thumbs-up icon on that specific icon of expectation and provided a verbal reinforcement such as "Great job for standing up by yourself," "Well done!" When the target child completed all the expectations, which usually was the time that the team line started walking to the school gate, the peer model handed in the visual cue card back to the researchers. At the same time, the researcher provided the feedback sheet to Henry's peers and briefly summarized their performance. There is no performance feedback sheet for the other six peer models. After this, the session ended.

Interobserver Agreement (IOA)

Interobserver agreement (IOA) was measured to ensure the reliability and believability of the coding system. The most often used index of believability that researchers convince themselves and others the data are trustworthy and deserve interpretation in applied behavior analysis is the measures of IOA (Cooper et al., 2019). The first data collector and observer was the author. A graduate student was trained to be a second observer and data collector to collect the baseline and intervention data. Then those data were compared with the one taken from the author to confirm whether the repeated measurement of the same event yields the same values (Cooper et al., 2019). The IOA was determined by using trial-by-trial IOA, which is a more conservative and meaningful index of IOA for discrete trial data, calculated by the following formula: Number of trials (items) agreement/Total number of trials (items) × 100 = trial-by-trial IOA % (Cooper et al., 2019).

Procedural Fidelity for Researchers

In order to determine whether all phases and conditions of this experiment were administered consistently and in accordance with the plan, the procedural fidelity data for researchers was collected in this study throughout the baseline, peer training session, and intervention for a minimum of 20% of those sessions (Cooper et al., 2019). The second observer collected the procedural fidelity data of the author. The author also took the procedural fidelity data on the second observer. During the peer training session, the second observer stayed around 10 feet away from the author for observation and data collection. During some of the baseline and intervention sessions, both observers stayed 5 feet from the peer model at a different spot and there was no discussion about data collection at this point.

Treatment Integrity for Peer Models

In order to ensure that the PMI program was applied exactly as planned and no other unplanned variables was administered inadvertently, treatment integrity data for peer models providing prompts and reinforcements was collected (Cooper et al., 2019) using the coding system described earlier.

Social Validity

Social validity is important to be measured for understanding opinions from the people who receive and implement the

intervention procedure (Owen-DeSchryver et al., 2008; Luiselli & Reed, 2011). In this study, the teachers, target children, and peer models' social validity were collected to appreciate their acceptability of and satisfaction with this PMI study procedure by using questionnaires and anecdotal records (Travers & Carter, 2021). The teachers and participants received questionnaires containing four to six closed- and open-ended questions to provide qualitative information (Kent et al., 2020).

There were four questions in the target children's social validity questionnaire: (1) Do you like [peer's name] to help you pack the schoolbag, lineup, and go home? Why? (2) Do you feel it is easier to finish the go-home routine when there is a peer helping you? Why? (3) Do you want a peer to help you with the go-home routine in the future? Why? (4) Anything else you would like to say? Similarly, there were four questions in the peer models' social validity questionnaire: (1) How do you like to be a peer helper, helping [target child's name] to pack up school bags and prepare to go home? Why? (2) In the future, if there is a chance, would you like to keep helping [target child's name]? Why? (3) In the future, would you like to help other friends to pack up their schoolbags and prepare to go home? Why? (4) Anything else you would like to say?

For the teacher's social validity questionnaire, there were six questions: (1) What changes, if any, did you see in your student's behavior as a result of their participation in this study? (2) Did your students develop any new skills as a result of their participation in this study? If yes, please describe these. (3) Did you see these skills used outside of the classroom? If yes, please describe these. (4) Would you suggest that this study be implemented with other children? If so, why? (5) Do you see yourself using peer-mediated intervention in the future? How do you imagine, if at all, using the peer-mediated intervention that was made for your students? (6) Is there any other feedback that you would like to share with us?

Due to the pandemic outbreak in Taiwan, the school closed, and students were all staying at home. The researchers were not able to deliver a paper-based social validity form to the teachers and participants in person. The teacher's social validity data were collected via email with a Word document. The teachers helped to email the social validity form to the child participants' parents, and the parents emailed their responses to the researchers in a Word document or video recording.

Result

Research Question 1: Effect of PMI on Target Children's Behavior of Completing Routines

Figure 1 is a dual-axis graph where the left y-axis shows the percentage of correct steps the target children completed, displayed using data points. The right y-axis shows how many specific steps the target children completed independently or with prompting, and this is displayed using bars. The dark grey bar means that the target child completed the expectation independently. The light grey bar means that the target child completed the expectation with a peer's prompt. The x-axis is the date.

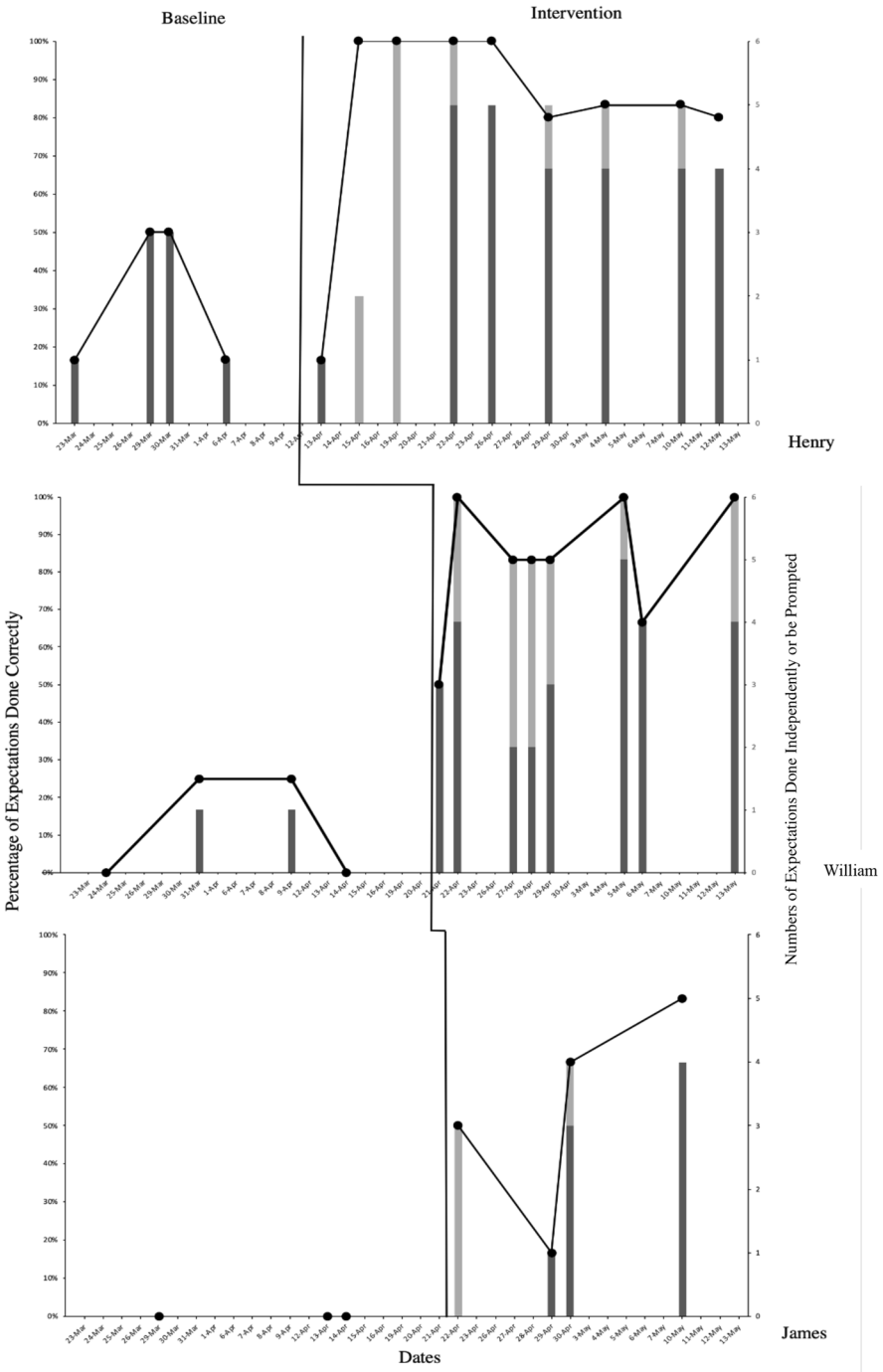
The effectiveness of the intervention was examined by visual inspection of the data. With the multiple probe design, the results show that behavior change did not occur prior to intervention. All three target children showed stability during baseline with steps completed ranging from 0 to 50% which were 0–3 steps. James had the lowest average percentage of expectations completed independently, which was 0%. Williams's average percentage of expectations completed independently was 15%. Henry had the highest average percentage of expectations completed independently, which was 33%. During the intervention, each target child demonstrated an immediate and substantial increase of expectations completed independently or with peer prompts, ranging from 16.65 to 100%. The average percentage of completed expectations for Henry, Williams, and James was 91%, 83%, 54%, respectively.

Research Question 2: Effect of Peer Training on Peer Model Supporting Target Children

Before the 40-min peer training, the peer models' baseline data were taken. Table 3 shows the pre-and post- of peer training data. All seven peer models in each of three target children groups shows 0% of correct implementing the prompting or reinforcement following each of six expectations. After the peer training, Henry's, Williams', and James' peer models correctly provided prompting averages were 77.5%, 97.5%, 100%, respectively. On the other hand, Henry's, Williams', and James' peer models correctly provided reinforcement averaged 66.67%, 95%, 100%, respectively.

Research Question 3: Treatment Integrity for Peer Models

After one 40-min peer training, James' peer models had provided prompts and reinforcement to each expectation correctly, so their treatment integrity was 100%. Williams' peer models' treatment integrity data were all 100% correct, except one day was 80% correct for providing prompting and 60% correct for providing reinforcement. Henry's peer models got 77.5% prompting correctly on average and 66.66% reinforcement correctly on average.



Note: X-axis is the date. The right y-axis is for the bar. The left y-axis is for the data point. Dark grey bar means that the target child did it independently. Light grey bar means that the target child did it with a peer's prompt.

Fig. 1 Data graph of target child's performance on completing the steps of go-home routine. X-axis is the date

Table 3 Peer model's pre-and post-training data in average

	Henry's peer models		William's peer models		James' peer models	
	Prompt (%)	Praise (%)	Prompt (%)	Praise (%)	Prompt (%)	Praise (%)
Pre	0	0	0	0	0	0
Post	77.5	66.67	97.5	95	100	100

Research Question 4: Social Validity

The result indicated that the target children who participated in the study overall enjoyed peers helping them during the transition time which was 100% agreement. All three of them felt it is easier to finish the transition with peers' help which was 100% agreement. All three of them would not mind getting help from the same peers in the future except James mentioned that he didn't like one of his peer models, so he didn't want his help in the future. Thus, 85.7% of peer models would be favored by the target children for continue receiving help in the future.

Five peers indicated that they like to help target children (i.e., 71.43% agreement), and two of them responded that they don't really like to help out for James. There were five peers who responded that they would like to keep helping the target children in the future if there is a chance (i.e., 71.43% agreement). All of the peers mentioned that they would like to help other friends with the go-home routine in the future (i.e., 100% agreement). Table 4 shows the full responses from the peer models.

Procedural Fidelity of Researchers

The procedural fidelity of the researcher was 100% which means the researchers consistently followed the procedures that were planned for all children. In this study, 33.33% of peer training sessions, and 21.875% of baseline and intervention sessions were collected for procedural fidelity.

Interobserver Agreement (IOA)

To this end, 21.875% of all sessions across baseline and intervention were randomly selected and coded independently by the main researcher and a well-trained graduate student. The percentage of agreement was 100% for both the peer models and target children's data.

Teachers' Social Validity Data

All three teachers were overall satisfied with the research process. Henry's and William's teachers reported that they felt the children had improved their independence in finishing the routines. Henry's teacher even reported his generalization of following expectations and showing independence outside of the research. They all agreed that they will think about using PMI in the future but there was not enough time for them to do systematic peer training. On the other hand, James' teacher mentioned that she hoped the intervention time could be longer to see more of the effectiveness of the PMI. She felt James was improving but not enough to the criteria that she set for her students in class.

Discussion

Research Question 1: Effect of PMI on Target Children's Behavior of Completing Routines

The findings of this study demonstrate that the PMI program was functionally related to target children's behaviors in an increasing percentage of correct routine steps done correctly. This result is consistent with the positive outcomes of previous PMI research (e.g., Katz & Girolametto, 2013; Thiemann-Bourque et al., 2017, 2018; Hu et al., 2018; Maich et al., 2018; Kemp et al., 2019; Boudreau et al., 2021; Dueñas et al., 2021a, 2021b), and extends those findings to teaching behavioral expectations of routines for students on the autism spectrum and student with ADHD. If there are more sessions and each of the sessions are with different peer models, it is suggested to report detailed information of which peer models are supported in which session. For example, Kemp et al. (2019) provided a table and reported specifically what each data point represents in terms of which peer models supported each target child on the autism spectrum during baseline, intervention, and maintenance phases.

Research Question 2: Effect of Peer Training on Peer Model Supporting Target Children

The data compared between pre- and post- training shows that a one-time 40-min peer training can overall effectively help peer models to prompt and reinforce the target children. The data of Henry's peer model is lower than other two target children's peer models because one of Henry's peer model was having a hard time to prompt Henry in a gentle way, refused to use visual card by putting cards away in her drawer, and failed to provide reinforcement after being prompted by the researcher. Therefore, the first four sessions of her data lower the average percentage of correct implementation for Henry's peer model. Even though the teacher referred a student as a potential peer model, the student did well during the peer training session, and the student passed the criteria when role-played with researcher, there is still a possibility that the student is not willing to use the skills that were taught. Future research is recommended to have more than one trained peer models and be flexible on switching the trained peer models in case some situation happens.

Research Question 3: Treatment Integrity for Peer Models

In this study, the treatment integrity data were reported as a group of peer models for each target child because there were not many sessions, and the data were overall 100%

Table 4 Table of peer models' full response to the social validity question

	Q1	Q2	Q3	Q4
P1(J)	Like it a little. Not very much. Because James was not listening on purpose. He was smiling and saying "NO". I like it because if we are slow, the teacher will take points out. But the teacher didn't because I am helping James.	No, because he was not listening.	I would like to give it a try. Helping others is a good thing.	Hope your research will be good!
P2(J)	I like it because I can help James finish packing up school bag and preparing to go home earlier.	Yes, because I can help James finish packing up school bag and preparing to go home earlier.	Yes, because it is a simple thing to help classmates.	To be honest, it's not a hard job to help classmates.
P3(J)	Soso… because James sometimes is compliant, but sometimes is not. If he is not compliant, it will delay a long time, and I will go home late.	If there are other children to help James with me, I will be willing to. Because in this way, there are people who rotate with me. However, I like to go to the rainbow room (i.e., resource room) with James during naptime.	Yes, but once a week only. Otherwise, I am a little bit tired.	Nothing.
P4(H)	I like it because it can help him to go home on time.	Yes, because it can help him.	Yes, because I like to help.	Go! Go!, Henry!
P5(H)	Yes, because this is my job.	Yes, because he is a considerate classmate.	Yes, because classmates should help each other.	Thank you for giving me this opportunity to experience.
P6(W)	I feel just like as usual helping other classmates. No special feeling. I feel good to help classmates who need help with.	If there is an opportunity, I would love to. Classmates should help each other.	Yes, if I can help, I would love to.	Nothing.
P7(W)	I like it because I like to help others.	I am willing to because William is my friend. Friends should help each other.	I am willing to. Because I have experience of helping William, I believe I can help other classmates without having any questions.	Happiness lies in helping others. I hope I can help someone who needs help with in the future.

J James' peer, *H* Henry's peer, *W* William's peer

correct for six out of seven peer models. Future studies can consider reporting the peer models' treatment integrity data in an individual way if there are more sessions or are a couple of different data between each peer model. Besides, the result of treatment integrity data for peer models means that a 7- to 8-year-old child is able to be a peer interventionist through adult support and training. Future research is recommended to collect peer models' generalization data across individuals by having them support other children who may also need help. It is also recommended to collect peer models' generalization data across locations by having peer models use the same behavior techniques in a different classroom. Collecting peer models' maintenance data on implementing behavioral techniques after the intervention ends for a while is recommended as well to identify whether peer models keep using those techniques even though intervention is ended.

Research Question 4: Social Validity

Results of the social validity questionnaires showed target children and peer models' overall satisfaction with the PMI program. However, there is social desirability bias (Travers & Carter, 2021) that the treatment integrity and anecdotal data of one of Henry's peer models did not correspond to her response in the social validity questionnaire. She indicated that she enjoyed helping Henry, appreciated the peer training, and would like to participate in PMI program in the future if there is a chance. Yet, her treatment integrity data showed low which was between 0 and 20% correct implementation of prompting and reinforcement. After the researcher's immediate prompt, she was still having a hard time following the steps that were trained in the peer training session. The performance feedback sheet was given to her at the end of the session. However, due to the time constraint, we did not have enough time to see the growth of her treatment integrity data. According to this peer model's social validity data, treatment integrity data, and anecdotal data, her perception and satisfaction of this PMI program were uncertain. This situation is resonant with the suggestion of Travers and Carter (2021) to use multiple aspects in collecting social validity data to understand participants' perceptions of participating in the study in broader domains to avoid social desirability bias. On the other hand, future research is recommended to implement a retrain session or provide not only one training session for the peer models if feasible, which was implemented in some of the previous studies (e.g., Katz & Girolametto, 2013, 2015; McCurdy & Cole, 2014; Lee & Lee, 2015; Thiemann-Bourque et al., 2017, 2018; Boudreau et al., 2021).

Although the overall feedback of peer models is positive, two peers who were both James' peer models reported that they did not really like to help James due to his non-compliant behaviors. Similarly, most peer models stated that they would like to help the target children in the future, but one of James's peer models pointed out that James was not listening so she wouldn't like to help him in the future. Another of James's peer models said that she would be more willing to help if there was another peer model who could support the target child with her and rotate the duty with her. Future research can consider having other interventions, such as providing a social story of routine steps and role play with the target child to offer an intervention package with PMI to support the target child who has more undesirable behaviors (Wahman & Anderson, 2021).

A similar situation occurred in the study of Dueñas et al. (2021a) that two peer models felt it was not hard at all to play with the target child at the beginning but changed to feel a little bit hard to play with the target child in the middle and at the end of the intervention. They further reported the reason is that their peers on the autism spectrum were not always sharing toys which are similar to what was mentioned that the target child's noncompliant behaviors affect the peer model's experience to help the target child.

James mentioned that he didn't like one of his peers so he didn't want his help; however, James still followed that peer's prompts and reinforcements. Interestingly, that peer reported enjoying helping James. Also, according to the researchers' observation and anecdotal data recording, the peer model showed responsibility and accuracy of treatment integrity in supporting James. Future research is suggested to keep collecting child participants' data to understand their perceptions. Especially when it comes to the friendship measurement, it is important to collect both target child's and peer model's behavior data and qualitative data to first make sure those two data show consistency and secondly confirm that it is a bidirectional friendship.

Limitation and Future Research

No Maintenance and Generalization Phases

Due to the pandemic outbreak, the school was closed. Researchers did not have the opportunity to take maintenance data to see if the target children can keep finishing the routines independently. Future research should set the same condition as a baseline after the intervention is ended for a period of time (e.g., 4 weeks) and take data to see if the target child maintains the skills. Another benefit to collecting maintenance data is to measure and observe if the target child and peer model form the beginning of their friendship by continuing to help the target child after the intervention ends (Harper et al., 2008).

Because we did not obtain consent from other subject instructors, we were unable to generalize our findings across other transition processes. Future studies should, if feasible,

examine generality across multiple contexts with comparable transition processes (e.g., transfer from the original classroom to a science classroom) (Chang & Locke, 2016). Future research can consider measuring the generalization across different peer models and collecting data on the target children's behavior in accepting assistance from different peer models (Chang & Locke, 2016; Dueñas et al., 2019).

Multiple Probe Design

In this study, we used multiple probe design because we had a limited number of researchers to collect data. On the other hand, we would like to decrease the number of times that the researchers walked into the classroom and interrupted the class. Additionally, it also decreased the possibility of observer effects that the target children might be aware of themselves being watched and changed their behaviors. Therefore, future studies may choose to use multiple probe designs when the extended baseline measurement may prove reactive, impractical, or costly (Cooper et al., 2019).

In our design, one of the limitations is that there is only one staggered date for starting the intervention between the second and third participants (i.e., Williams and James). The reason is that James's attendance started to be unstable and his parents were thinking about transferring to other schools, so we decided to start the intervention as soon as possible. Future studies should have more staggered dates or sessions if it is possible.

Multiple probe design is a type of single-subject design which involves a smaller number of participants than a group design. One of the strengths in using single-subject design is to be able to focus on each individual's differences and make adjustment according to the individual's situation. At the same time, this could be one of the limitations in using single-subject design that does not include a large group of participants. However, additional single case studies that replicate this study can contribute to further evidence supporting PMI.

Measurement

Another limitation is that the measurement of dependent variables in this study is the number/percentage of expectations implemented correctly by the target children which might not be able to completely present the effectiveness of the PMI program. For example, the latency of starting the expectations and the duration of completing the routines are decreasing. Future studies may use multiple measurement tools to add more details on the children's behavioral changes to identify the effectiveness of the intervention. A future study can measure the duration of finishing routines to see if the target children decrease their time spent on completing the routines. Besides, future research can measure the latency data which is the time between the teacher giving an announcement to start the routine and the target child starting to do the routine. The decreased amount of time in latency data during the intervention can help to identify the effectiveness of the intervention.

Procedural Fidelity Data

In this study, we reported IOA of procedural fidelity in a final percentage because it is 100%. However, if IOA is not reaching 100%, it is recommended to report the detailed information to increase the reliability of procedural fidelity. For instance, Kemp et al. (2019) reported in a table listing detailed information about which session was recorded IOA with every item (e.g., prompt peer to take turns or block incorrect turn) comparing results between the primary and secondary data collectors. Future study is recommended to collect procedural fidelity data to measure the effectiveness of PMI in a more accurate and reliable way (Schaefer et al., 2016).

Social Validity Data

The questionnaire with four to six close- and open-ended questions and anecdotal records was used to collect social validity data in this study, which was consistent with previous research suggestions from Kent et al. (2020) to gather qualitative information from peers to understand their personal benefit, enjoyment, motivation, and the appropriateness of this intervention and from Travers and Carter (2021) to use multiple aspects in social validity data collection. Three limitations of collecting social validity in this study are discussed. First, is the inconsistent way of social validity data collection. Due to the pandemic outbreak in Taiwan, the school closed, and students were all staying at home. The teacher's social validity data were collected via email with a Word document. The teachers helped to email the social validity form to the child participants' parents, and the parents email their responses to the researchers. Child participants answered in typing, video recording, or responding verbally, and then their parents typed down for them. Second, the parents might provide prompts or hints to the child participants to answer the question which may affect the child participants' answer. The researchers were not able to train parents on how to collect social validity data with a neutral tone because it is difficult to contact parents under a stay-home order. Third, the researcher could not ask the follow-up question and observe the children's facial expressions of their responses because the researchers were not able to deliver a paper-based questionnaire to the school teachers and child participants in person.

Implication for Practice

Peer Selection

Covey et al. (2021) suggested that practitioners pick peer models strategically based on student preferences, skill levels, or personalities, since this may yield better outcomes than random pairing. Peer qualities are one of the aspects that have previously been urged to be recorded and measured as prospective moderators to offer education practitioners with a reference on peer selection (Locke et al, 2012; Chang & Locke, 2016). Practitioners might assess if the child is popular, socially visible, compliant, demonstrating leadership potential, having strong problem-solving abilities, having influence, and having self-confidence, all of which are highly connected with peer selection (Jackson & Campbell, 2009). Similarly, Locke et al. (2012) found that teachers frequently organize peer models that are more socially adept and have a higher social position in the class. Here are some more issues to consider for practitioners: If the child is enthusiastic during interactions with other children, if the child is able to make up schoolwork missed during training, is well-liked by typical peers/other peers, is a positive social partner with the target child, is able to do the target skills or activities to certain criteria, has willingness to participate, attend school on a regular basis (Harper et al, 2008; Trembath et al., 2009; Dueñas et al., 2021a, 2021b).

Include More Than One Peer Model

Practitioners may select more than one peer model for each target child on the autism spectrum. According to the social validity data in this study, two of the peer models mentioned they felt it was good that there were other peer models rotated with them, so they were more willing to participate in the PMI program. Harper et al. (2008) mentioned that having a group of students supporting one child with autism may provide a better ratio and increase the typical peers' motivation. Similarly, according to one of the peer model's social validity data in this study, she mentioned that if there are other peers to help target children with her, she would be more willing to help out. On the other hand, educators can consider engaging the entirety of peers who stay with the child with disabilities in the same classroom to create a positive and supportive learning environment (Trembath et al., 2009; Katz & Girolametto, 2013, 2015; Gunning et al., 2019).

Peer Training

Practitioners are encouraged to use behavioral skills training which contains an introduction, modeling, rehearsal, and immediate feedback to provide peers with systematic training (Covey et al, 2021). Without structured peer training, practitioners may find it hard to see a significant increase in the abilities of children on the autism spectrum because the peer models are less likely to know how to support children with disabilities in an effective way (Chang & Locke, 2016). According to the teachers' social validity data in this study, some teachers are too busy to provide systematic peer training. In this case, the teachers are encouraged to cut the steps or expectations into small pieces to teach peer models one target step at a time in a small amount of time (e.g., 5–10 min) during the break time, nap time, or playtime. Additionally, teachers can train peer models in groups to increase the efficiency of training and save teachers' time spent on peer training (Covey et al., 2021).

Focusing on Peer Models

Educators who implement the PMI program should collect data on all involved students including children with and without disabilities by using multiple measurement tools such as anecdotal data, close-ended survey, direct observation, open-ended survey, interview, or written reflection (Travers & Carter, 2021). Practitioners should listen to child participants' perspectives and keep observing and monitoring the progress of the PMI program to make the adjustment accordingly. For example, in this study, the researcher asked the peer models whether they would like to be trained in the resource room during nap time before starting the peer training. Moreover, the researcher told the peer models that they can choose not to help the children with disabilities at any time without penalty. Receiving quick verbal consent and providing the options can show that the teachers are not only focusing on helping the students with disabilities but also caring about the willingness of the peer models. This resolves the critics from some researchers stating that the PMI program might cause peer models to receive less teacher attention and decreased learning opportunities (Ferraioli & Harris, 2011).

Conclusions

Our work increases the use of PMI in teaching the behavioral expectations of routines, which is the self-regulation skill for students with ADHD and on the autism spectrum, by using a multiple probe design. Second, because there are few studies concentrating on PMI in Taiwan, our study expands the number of PMI studies based on the ABA principle and enhances the emphasis of PMI in Taiwan. Third, our study gathered treatment integrity and social validity data from peer models in accordance with the suggestions and gaps identified in prior studies concentrating on peers without disabilities (Travers & Carter, 2021). Fourth, our study focuses

on children in inclusive school settings, which is consistent with past research that supports assessing the efficacy of PMI for peer models and focusing on students who are closer in age (Covey et al., 2021). Furthermore, it aligns to the Taiwan Special Education Act (2019) on inclusion, Taiwanese indicators of inclusive education (Huang et al., 2017), and the present trend of settling for children with disabilities (Taiwan Ministry of Education, 2020). Finally, our study strategically matched students and suggests that future research should pick peer models in a more methodical and data-driven manner. To summarize, the PMI program is an evidence-based strategy with no detrimental effects identified in studies (Schaefer et al., 2016; Travers & Carter, 2021), so future research is encouraged to replicate the PMI program to extend its usage to more diverse participants and settings and report the data on all participants who are in the PMI program.

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Declarations

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